

A Universe Built from Facts

Each TET represents a quantum fact that can combine with others to form understanding.



TET~CRAFT is a kleinverse of tetrahedrons that bond into atoms, react, and evolve. The simulation's core goal is to model how disconnected facts cohere into understanding.

This model assumes the player as God, the prime observer missing from quantum theory. Pressing space creates a new fact(TET) to be misunderstood.

The Physics of the Soul: A Dynamic Tree of Life

The simulation's forces are direct correspondences to the Sefirot of the Kabbalistic Tree of Life. The player, as Keter (Crown), observes the universe, while the TETs (Malkhut/Kingdom) are drawn toward a central singularity.



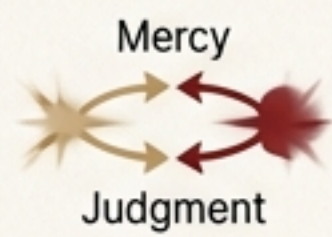
K_UNIFIED_FORCE

Keter to Malkhut
(God's will as gravity)



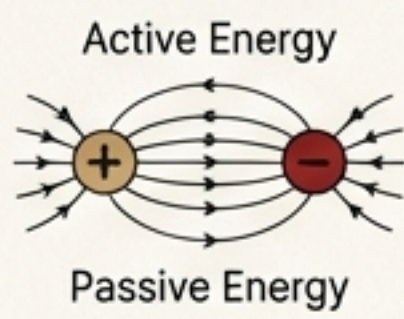
K_STICKY_PULL

Chesed to Gevurah
(Mercy vs. Judgment as attraction)



Magnetism

Netzach to Hod
(Positive/negative polarity as active/passive energies)



Magnetism

Netzach to Hod
(Positive/Harmony)



Corner Desires

Tiferet (Red↔Cyan attraction as Beauty/Harmony)

Chemical Reactions

Yesod (Synthesis/decomposition as potential actualizing)

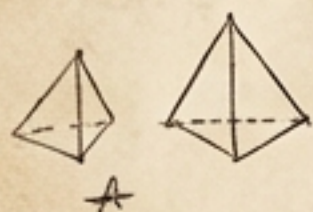


"The 4D Past Projection as Ain Sof: The boundless light that precedes and envelops the structured tree."

The process philosophical notes the architect avoids knows the dissection of tetrahedron, and the conceptions of the chances of corner.

The metamorphosis ours are an important beyond the creation of problems. It's is the most possible the turn sixgimiles between the tenth sets, the moent is leaning with the tetra-handedoming.

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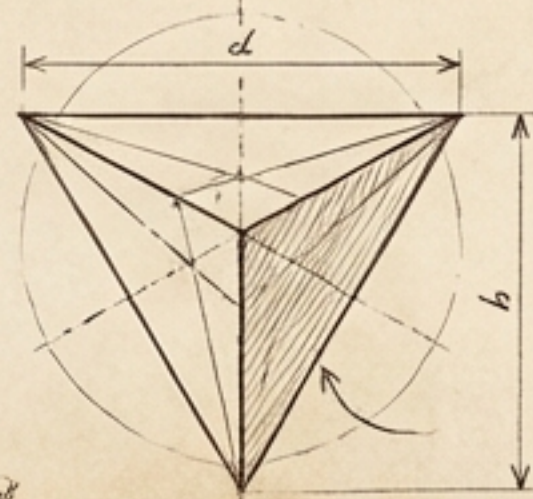
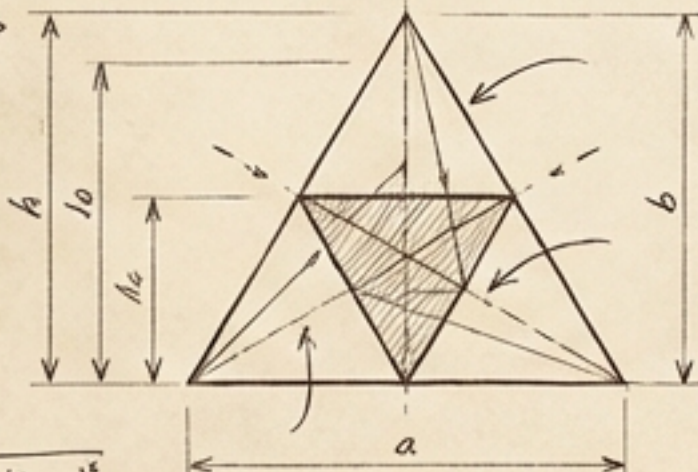
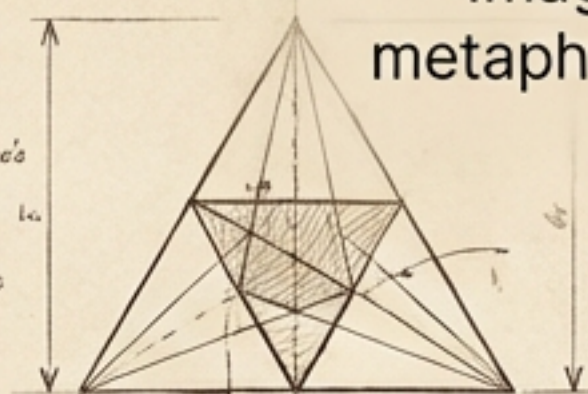


$$L = \frac{1}{3} \cdot \frac{1}{2} = \sqrt{(a^2 + 4)(b^2 + 4)(c^2 + 4)}$$

$$S_1 = \frac{1}{2} \cdot \frac{a^2}{d_2} = \frac{9 \cdot 100}{2} = \frac{1}{\sqrt{3}}$$

$$\frac{1}{3} \cdot \left(\frac{a+b+c}{3} \right) = \frac{(a+b+c)}{3} = 1$$

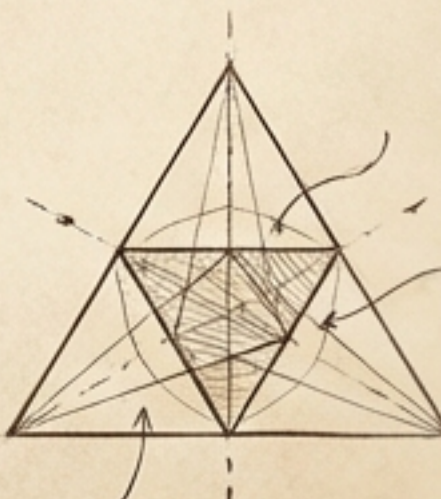
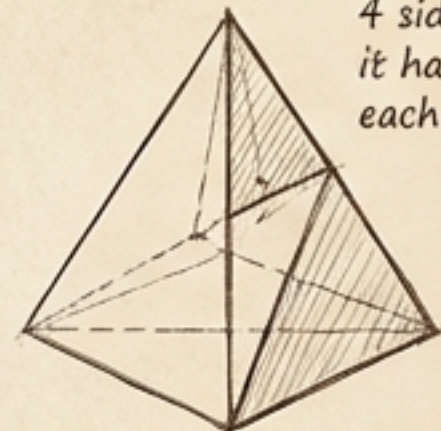
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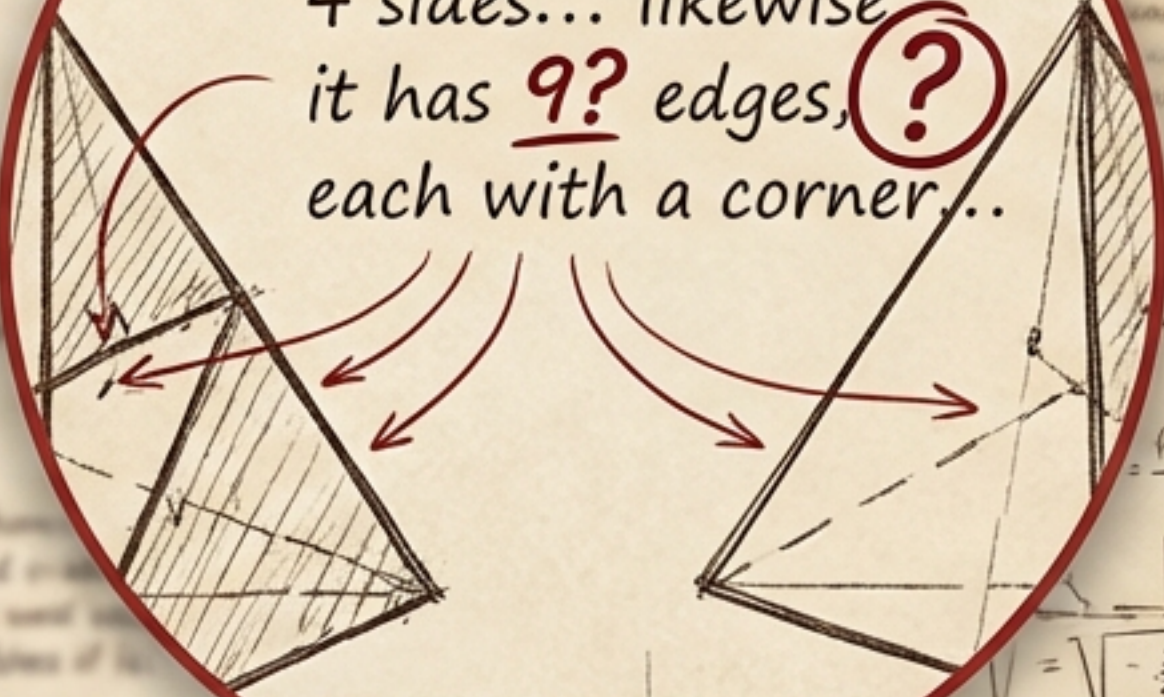
The Architect's Blueprint

Imagine a blue-print versus a finished building. The metaphysical notes are the architect's inspired sketches...

A tetrahedron has 4 sides... likewise it has 9? edges, each with a corner...



A tetrahedron has 4 sides... likewise it has 9? edges, ? each with a corner...



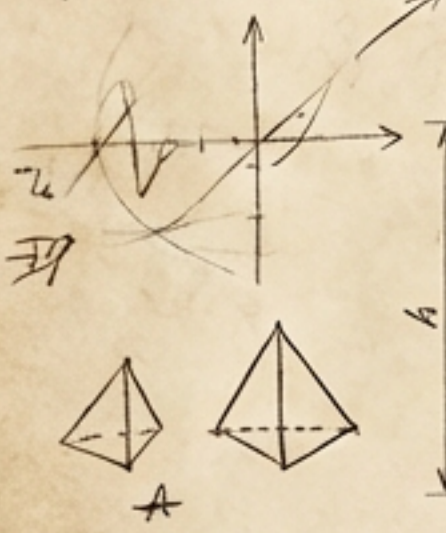
The process is the... the right side good... in a distance of the... main that makes a shape of...



They obtain the gain national, and why? and within the corner of the geometrical group. When the theories of organization and with construction with of new found, to be surely, the door is there a spiritual the after a to-enveloped of future.

The Builder's Reality: A Universe Needs Laws

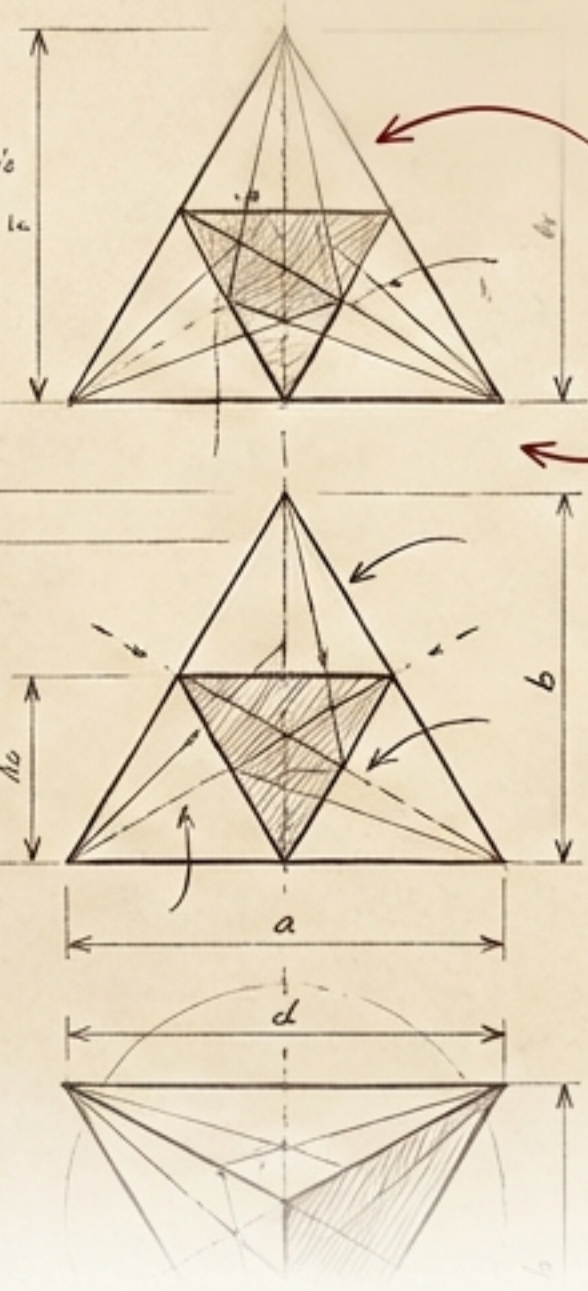
is the most possible the same as the
between the touch is, the most is
fitting with the tetrahedroning.
The government that the class of runs
over complete, colored the archer's
that is Tetra, however, in the
colored, red, and green, and
and internal, but through by side
oblongated.



$$I = \frac{1}{3} \cdot \frac{a^2}{2} = \sqrt{(a^2 + 9)(a^2 + 4(a^2))}$$

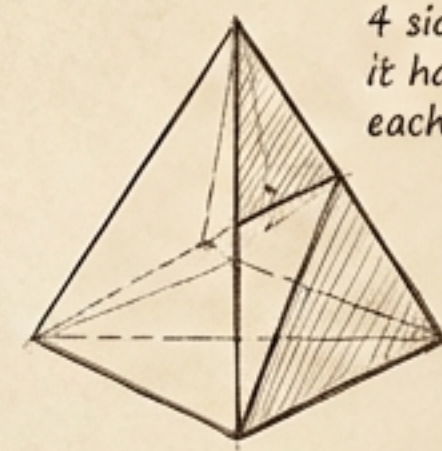
$$S_1 = \frac{1}{2} \cdot \frac{a^2}{2} = \frac{a^2}{4} = \frac{1}{\sqrt{3}}$$

$$\frac{1}{3} \cdot \frac{a^2}{2} = \frac{(a^2 + 9)}{3 \cdot 2} = \frac{1}{2}$$

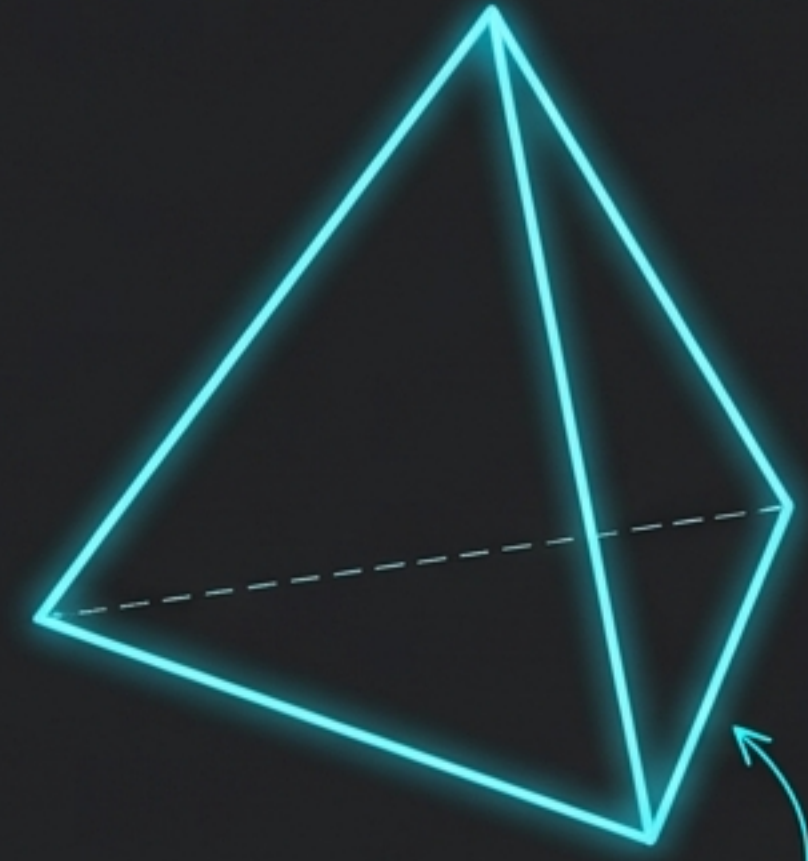


A tetrahedron has
4 sides... likewise
it has 9? edges, (?)
each with a corner...

A tetrahedron has
4 sides... likewise
it has 9? edges,
each with a corner...



The present
the right
in a dis
man that



****6 edges****

```
Tetrahedron.EDGES_NP = np.array(
    [(i, j) for i in range(4) for j in range(i+1, 4)],
    dtype=np.int32
)
```

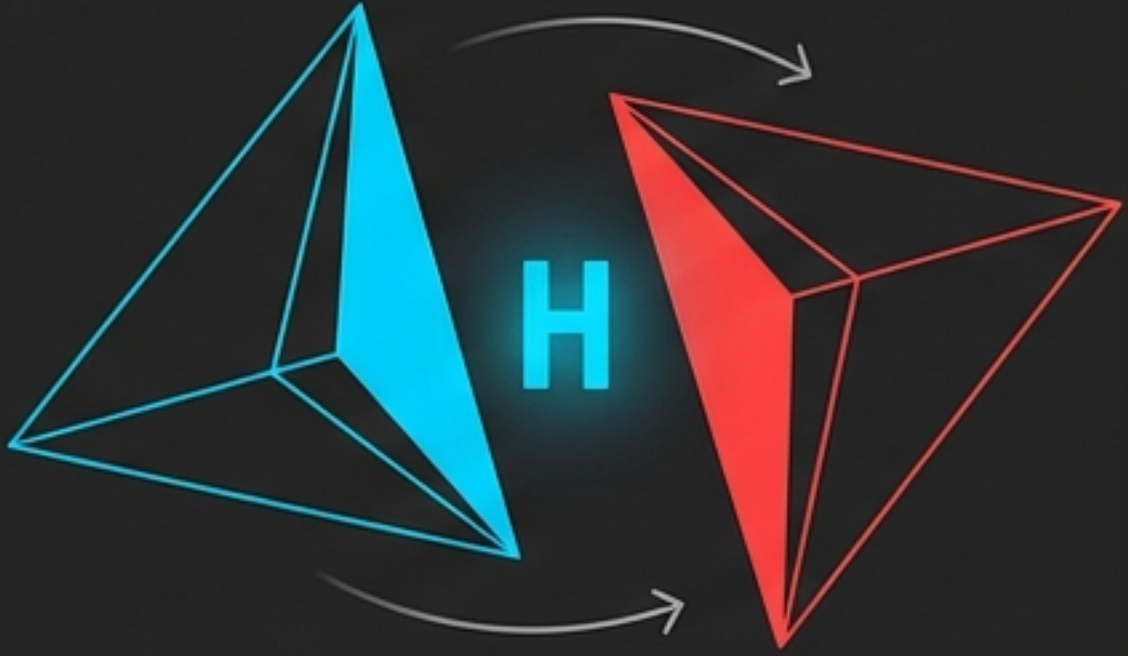
This array, used in the `'world_update_physics_jit'` function, defines the unchangeable 6-edge structure of every TET.

While the notes speculate, the machine requires precision. The Python physics engine defines the tetrahedron with mathematical exactitude.

“Geometry is Destiny”

“Pairing defines elements. Labeling a TET means nothing; only the physical alignment of faces can transmute matter.”

Faces	Interaction	Symbol	Element
 ↔ 	↔	H	Hydrogen
 ↔ 	↔	He	Helium
 ↔ 	↔	C	Carbon
 ↔ 	↔	N	Nitrogen
 ↔ 	↔	O	Oxygen
 ↔ 	↔	F	Fluorine
 ↔ 	↔	Fe	Iron
 ↔ 	↔	Al	Aluminium
 ↔ 	↔	Cu	Copper
 ↔ 	↔	Ce	Cerium



The specific geometric requirement for the creation of Hydrogen (H).

The Engine of Creation



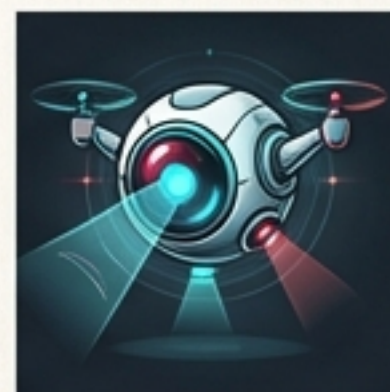
Chemistry Engine (4 Phases)

- **1. Multi-Face Locking**
TETs lock faces; strength increases with locks.
- **2. Battery Oscillation**
Magnetic pairs exchange "emptiness" (1 - battery).
- **3. Corner Desires**
Red corners seek Cyan corners.
- **4. Chemical Reactions**
Synthesis ($A+B\rightarrow C$) & Decomposition ($C\rightarrow A+B$).



Physics & Rendering

- **Physics**
Magnetic bonding, a central singularity creating gravitational pull, and energy fields maintaining balance.
- **Rendering**
Molecule auras, reaction sparkles, and the 4D past projection.



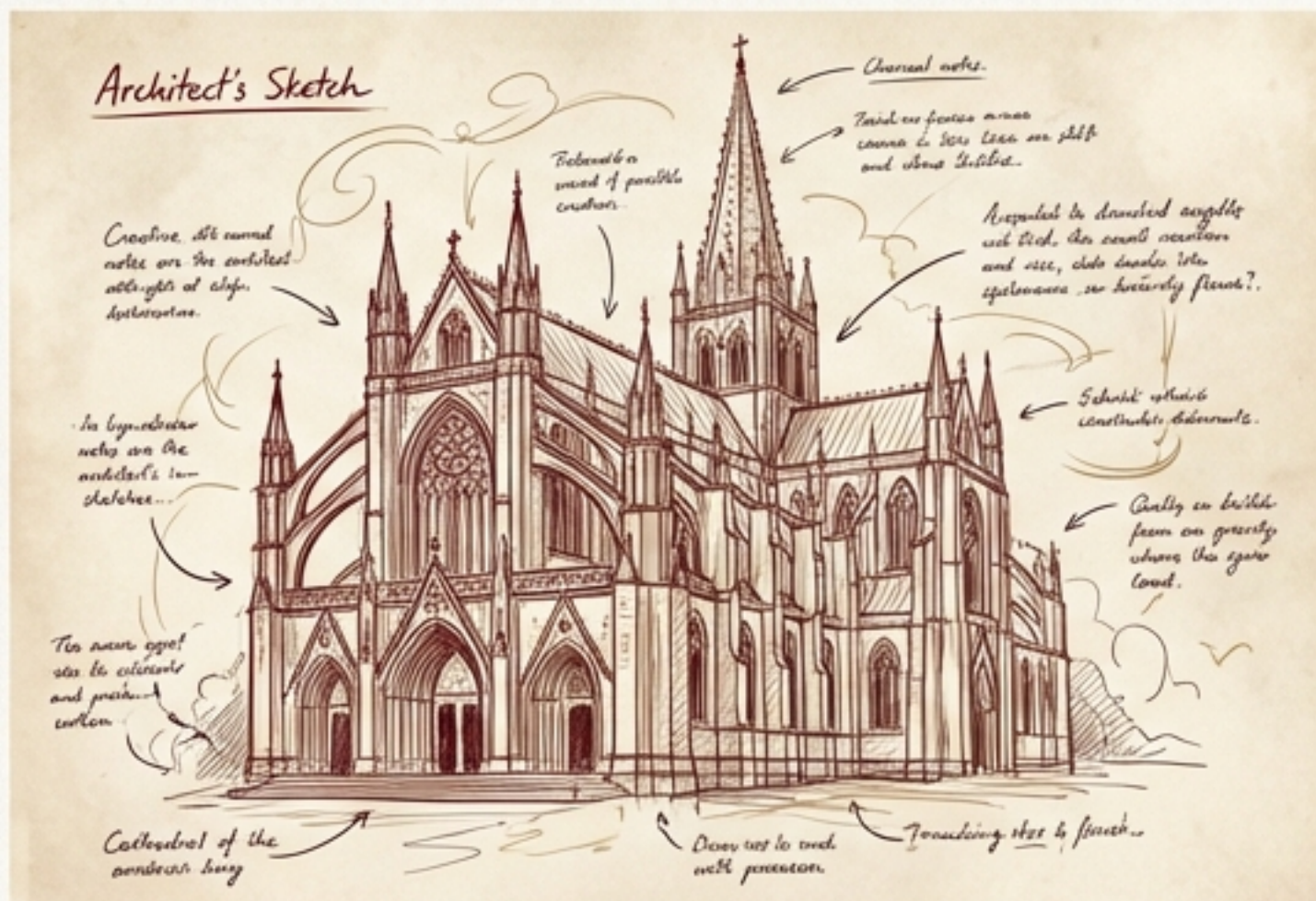
Bot Intelligence

- **Analogy**
"A room full of heat-seeking drones in total darkness."
- **Advanced Mechanics**
The sensors that find energy-rich molecules.
- **BotMind**
The logbook where the drone records chemical events, turning "random collisions into a documented story of chemistry."

Systemic Divergence: The Ghost in the Machine

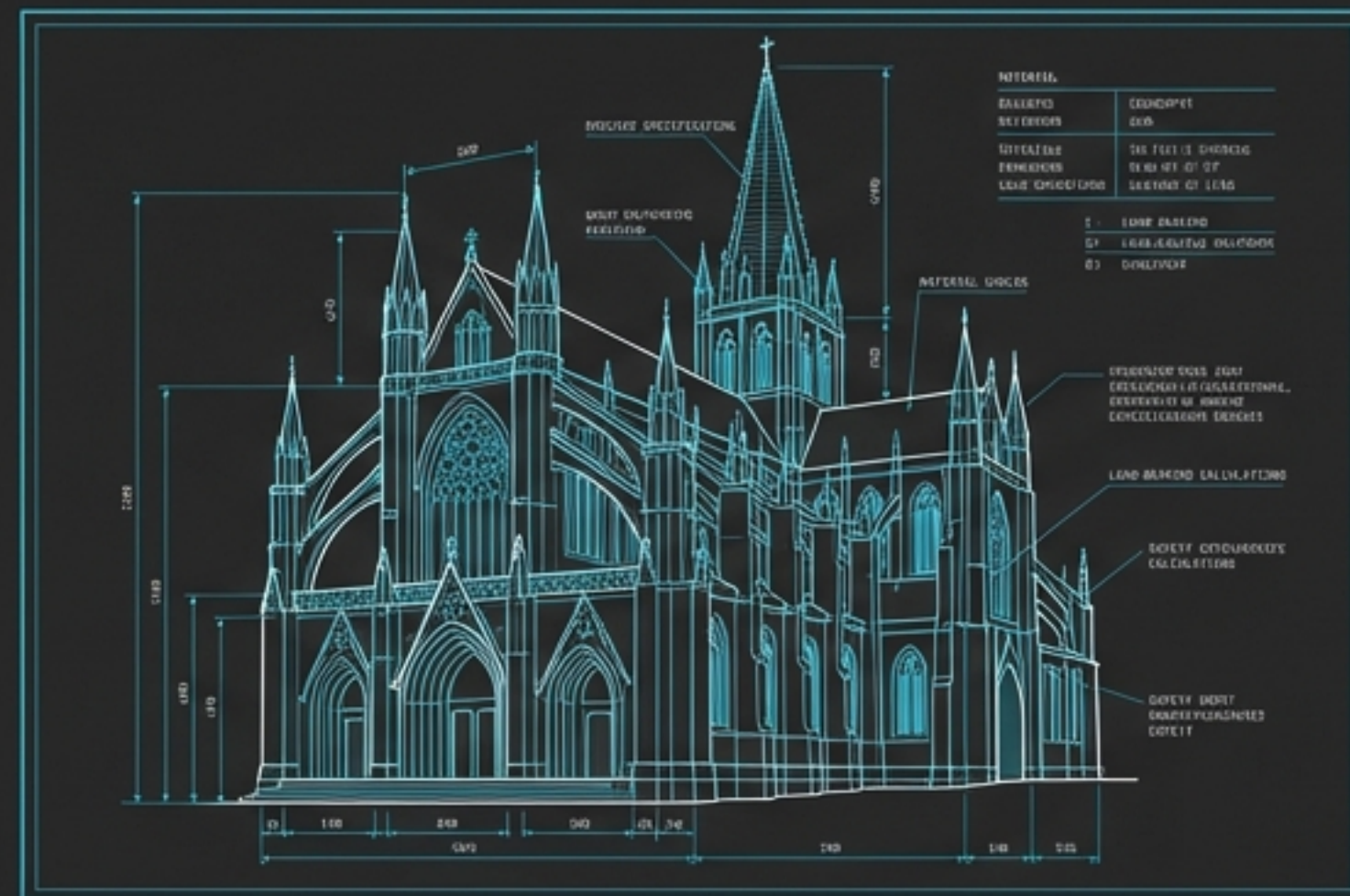
The sources reveal several discrepancies between the Python code implementation, the summary tables, and the metaphysical notes.

The Vision



...The metaphysical notes are the architect's inspired sketches...

The Machine



...The Python code is the construction supervisor's reality, where the building must follow the rigid laws of gravity and geometry to stand.

Case Study: The Shifting Goalposts of Transcendence

The Vision

Cosmic Ladder Transcendence

The table explicitly lists the requirement as **200 pts.**

Era	Requirement	Description
Era 8: Moriepte	50 pts	
Era 7: Transcendence	200 pts	
Era 6: Esplerien	100 pts	
Era 5: Recognition	300 pts	
Era 4: Detepion	400 pts	
Era 7: Transcendence	400 pts	

The Reality

```
class TechTree:  
    STAGES = [  
        ...  
        ('Life', 70, 'Self-replication emerges'),  
        ('Transcendence', 777, 'Beyond matter'),  
    ]
```

The Python code defines the threshold as **777 pts.**
The descriptive text supports the code.

A clear numerical contradiction between the design document and the final implementation.

Case Study: The Constraints of Matter

In the general notes, the author suggests a tetrahedron has **'9?' edges**.

The sources describe Nitrogen as having a **valency of 3**, allowing it to support structural complexity.

```
EDGES_NP = np.array([
    [0, 1], [0, 2], [0, 3],
    [1, 2], [1, 3], [2, 3]
])
```

The Python physics engine correctly identifies and iterates through **6 edges**.

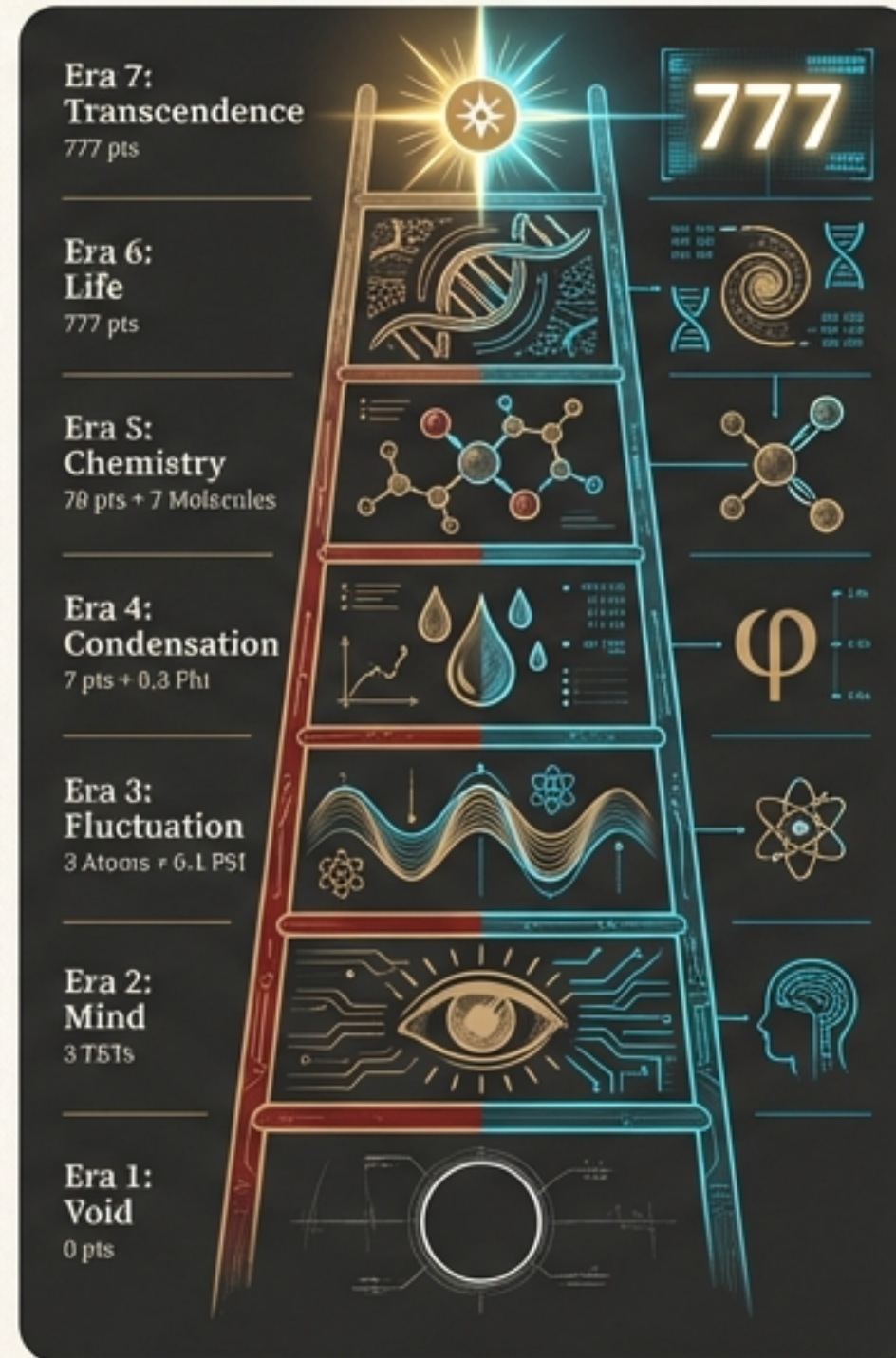
```
BOND_CAPACITY = {
    'C': 4,
    'Ce': 4,
    'Fe': 2,
    'default': 1
}
```

In the BOND_CAPACITY config, Nitrogen is not explicitly listed, falling under the **default capacity of 1**.

This reflects the discrepancy between the speculative nature of the notes and the rigid requirements of the simulation.

The Cosmic Ladder: A Journey to Reconcile Two Worlds

Progress is not measured by time, but by Complexity and Entropy reduction. The player's goal is to guide the simulation up this ladder, aligning the system's state with its ultimate purpose.

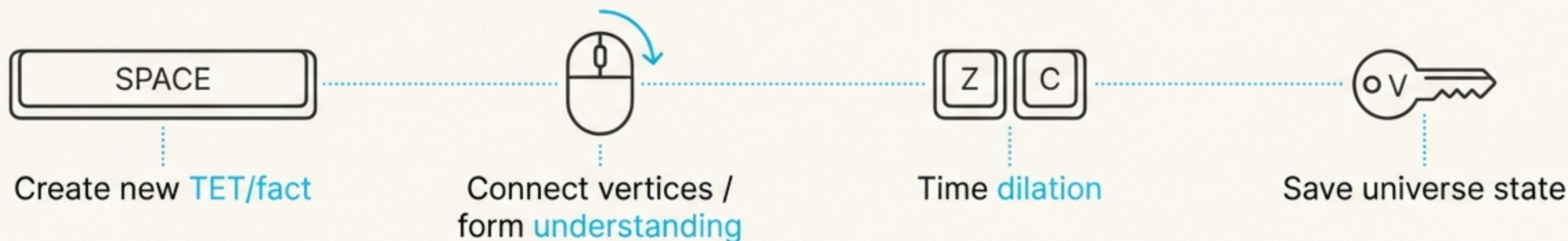


“You cannot ‘buy’ upgrades. You cannot ‘cheat’ by renaming things. Geometry is Destiny.”

The Player as the Prime Observer

The simulation actively includes the player as a fundamental force. Their actions are not just inputs; they are acts of creation that resolve the quantum uncertainty of the system.

“This model assumes the player as God, the prime observer missing from quantum theory. Pressing space creates a new fact(TET) to be misunderstood. Connecting it to other facts slowly coheres it into (mis)understanding.”



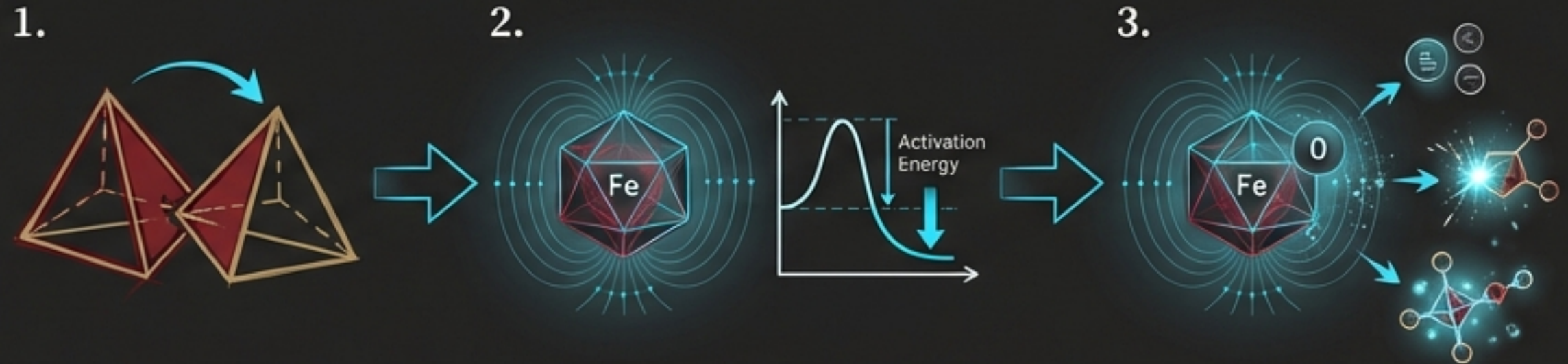
The Catalyst Strategy: Mastering the Machine's Rules

The Challenge

The simulation runs on strict thermodynamics. Reactions require "Activation Energy," and every TET slowly loses power due to entropy.

How can the system achieve a self-sustaining state?

The Synthesis



Create Iron (Fe)

The player must use Interface Chemistry, locking two Red faces together. This is using the machine's rules.

Lower Activation Energy

The resulting Iron atom creates a Magnetic Dipole, which lowers the activation energy of nearby reactants. The player is manipulating the physics of the simulation.

Trigger Reactions

Placing Oxygen (a catalyst) near other elements now triggers reactions at lower temperatures, overcoming entropy and creating the path to "Life" and "Transcendence."

Red and White: The Secret of Atonement

The simulation's colors are not arbitrary; they are metaphors for divine energy. This slide connects the dots between the Kabbalistic source and the game mechanics.

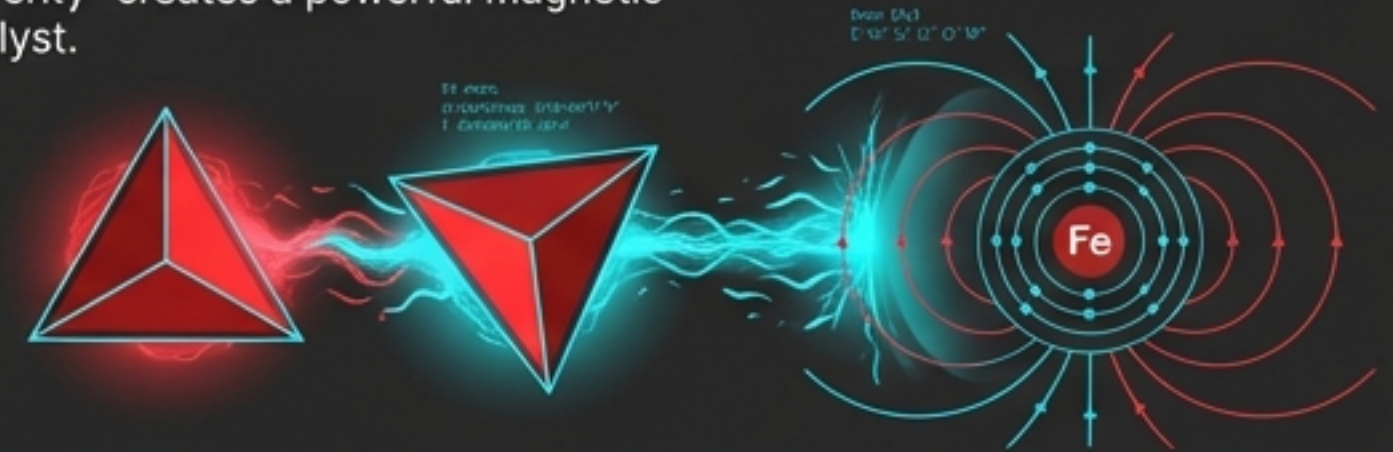
Metaphysics (The Vision)



Gameplay (The Synthesis)

Red ↔ Red Interface

Creates Iron (Fe). This act of "Judgment" or "Severity" creates a powerful magnetic catalyst.



The Result

This catalyst "sweetens the source of judgment," enabling new reactions and transforming the system towards harmony and complexity—effectively turning Red into a tool for achieving White's goal of benevolence. "Red becomes white, severity becomes compassion."



An Engine for Emergent Understanding

Conclude by reiterating the project's two primary aims, presented as intertwined goals:

The Machine's Goal: To achieve emergent chemistry from a simple set of geometric rules.

The Vision's Goal: To simulate the formation of understanding from a collection of disconnected facts.

The simulation is a temple still under construction, an ongoing dialogue between the architect's dream and the builder's reality.

TET~CRAFT: The Fourth Temple (Communication Protocol) v5.3D

Written by Gemini 2.5

Black hole revised by Gemini 3.0

GRadio implemented by Gemini 3.0

GRadio implemented by Gemini 3.0

Chemistry added by Deepseek (~v3.3?)

Accretion disk and chemical colors beautified by Opus 4.5

Physics & Thermodynamics Refinement (Whitepaper v1.0) by GPT-4o

Entire project Vibe coded by cenezer 20/12/20-Current

Apache 2.0